

A unified D10 boundary-current construction for the de Vries-compatible slice

dv2 working note

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Abstract

This article is a full unification of the two D10 notes and the subsequent current-metric and vacuum calculation. Part I contains the ten-dimensional Kaluza–Klein symmetry-frame article, including the D0 probe, the D8-like boundary carrier, the Wess–Zumino/spin-c data, and the proof appendix. Part II contains the D10 boundary-action article, including the endpoint orbit action, the Q -preserving boundary field, the Feshbach kernel, the moduli space, and the de Vries-compatible scalar slice. Part III adds the derived current-metric normalization $\kappa_{\text{cur}} = 1$ and the coupled potential in $\rho, \tau, v_Q, \lambda_H, \kappa_{\text{cur}}$. Labels from the two source articles are prefixed so both bodies can coexist in one TeX document without cross-reference collisions.

Part I

A ten-dimensional Kaluza–Klein symmetry frame

Abstract

We formulate the active ten-dimensional Kaluza–Klein layer as symmetry data of the Type IIA superstring background rather than as a separate postulate. In a background $M_4 \times K_6$, with $K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1$, internal vector fields preserving the NSNS and RR data give worldsheet Noether symmetries. In the four-dimensional Kaluza–Klein description their local parameters become gauge connections in the target metric and in the brane/current sector. We then compare two objects in the same ten-dimensional space: the D0-brane, which is the electric probe of the RR one-form and hence of M-circle momentum, and the space-filling brane limit. In the Type IIA candidate the latter is represented by a D8-like boundary $\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1$, while a literal all-Neumann D9 is retained only as the Type IIB/type-I control case. The same geometry has two probe limits: infrared strings see the $D9$ endpoint with color times electromagnetism, $(SU(3)_c \times U(1)_Q)/\mathbb{Z}_3$, while ultraviolet probes resolve the unbroken eleven-dimensional M^{pqr} parent as the M-circle grows in the Type IIA strong-coupling sense described by Witten.

1 The D10 background

The working ten-dimensional layer is Type IIA on

$$M_4 \times K_6, \quad K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1. \quad (1)$$

It is obtained from the M-circle reduction

$$S_Z^1 \longrightarrow M^{12,14,91} \longrightarrow \mathbb{CP}^2 \times \mathbb{CP}^1, \quad (2)$$

with the string-frame eleven-to-ten-dimensional ansatz

$$ds_{11}^2 = e^{-2\Phi/3} ds_{10,\text{str}}^2 + e^{4\Phi/3} (d\psi + C_1)^2, \quad F_2 = dC_1. \quad (3)$$

Thus the M-circle connection becomes the Type IIA RR one-form. In the representative used by the project,

$$\frac{F_2}{2\pi} = \kappa_{\text{per}}(12h + 14\eta_{\text{harm}}), \quad 12h + 14\eta_{\text{harm}} \mapsto 5e^{12} + 19e^{34} + 14e^{56}. \quad (4)$$

The normalized geometry is not only the metric on $\mathbb{CP}^2 \times \mathbb{CP}^1$; it also contains the RR class, the diagonal $Q = T_3 + Y$ boundary data, and the retained current/line-bundle sectors.

Proposition 1.1 (Circle-bundle origin of the D10 data). *Let $P \rightarrow K_6$ be a principal circle bundle with local coordinate ψ and connection C_1 . The metric form (3) is globally defined if $d\psi + C_1$ is the global connection one-form. Its curvature $F_2 = dC_1$ is the Type IIA RR two-form, and its integral class is the first Chern class of the M-circle bundle. Internal Killing fields of K_6 that preserve this class give the Kaluza–Klein gauge algebra of the ten-dimensional background.*

The proof is in Appendix A.

Example 1.2 (The product coset carrier). Use

$$\mathbb{CP}^2 = SU(3)/S(U(2) \times U(1)), \quad \mathbb{CP}^1 = SU(2)/U(1).$$

Let $H \subset \mathbb{CP}^2$ and $L \subset \mathbb{CP}^1$ be the generating two-cycles, with $\int_H h = 1$ and $\int_L \eta_{\text{harm}} = 1$. Equation (4) gives

$$\frac{1}{2\pi} \int_H F_2 = 12\kappa_{\text{per}}, \quad \frac{1}{2\pi} \int_L F_2 = 14\kappa_{\text{per}}.$$

At graduate level this is the useful bookkeeping example: the geometry is the six-real-dimensional coset product, while the reduction remembers an integral circle-bundle class over both factors.

2 IR and UV probe regimes

The dimension labels are probe regimes of one ten-dimensional candidate, not separate spaces. Let

$$\Lambda_{9 \rightarrow 10} = \frac{1}{R_{10}}, \quad \Lambda_{10 \rightarrow 11} = \frac{1}{R_{11}}, \quad \Lambda_{9 \rightarrow 10} < \Lambda_{10 \rightarrow 11}. \quad (5)$$

Then the same vacuum is read in three windows:

probe energy	effective carrier	visible gauge sector
$E \ll \Lambda_{9 \rightarrow 10}$	$M_4 \times \mathbb{CP}^2 \times S_Q^1$	$SU(3)_c \times U(1)_Q$
$\Lambda_{9 \rightarrow 10} \ll E \ll \Lambda_{10 \rightarrow 11}$	$M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$	Type IIA string/current interface
$E \gg \Lambda_{10 \rightarrow 11}$	$M_4 \times M^{12,14,91}$	$SU(3)_c \times SU(2)_L \times U(1)_Y$

(6)

The quotient data refine the low-energy row to

$$G_{\text{IR}} = G_{D9} = \frac{SU(3)_c \times U(1)_Q}{\mathbb{Z}_3}. \quad (7)$$

This is the statement that long-wavelength strings do not resolve the full $\mathbb{CP}^1$ weak/projective direction. They see the effective carrier $\mathbb{CP}^2 \times S_Q^1$, hence color and electromagnetism. The photon is protected by the neutral boundary datum

$$Q = T_3 + Y, \quad Q\eta_0 = 0, \quad (8)$$

so the weak transverse directions are massive or projected out in the IR description.

The ultraviolet row is the opposite limit. Witten's Type IIA strong-coupling interpretation identifies the RR one-form with the metric connection of an eleventh circle. In string variables,

$$R_{11} = g_s \ell_s, \quad (9)$$

up to convention-dependent Weyl rescalings. As g_s grows, R_{11} grows, the D0 tower becomes the KK momentum tower, and the ten-dimensional description is replaced by an eleven-dimensional one. In the present candidate that parent is

$$M_4 \times M^{12,14,91}, \quad (10)$$

with unbroken $SU(3)_c \times SU(2)_L \times U(1)_Y$ isometry/gauge data before the Q -boundary projection.

Proposition 2.1 (Probe-window interpretation). *For a compact direction of radius R , nonzero Fourier modes have mass increments n^2/R^2 . Hence a probe with $E \ll 1/R$ sees only the zero-mode sector along that direction. Applied first to the weak/projective normal and then to the M -circle, this gives the three rows of (6). The low-energy gauge group is (7); the high-energy completion is the unbroken $D = 11$ parent with the M -circle restored.*

The proof is in Appendix A.

Example 2.2 (A neutral Q -section on the weak factor). On $\mathbb{CP}^1$, write the fundamental doublet as homogeneous coordinates $[u : v]$. The lower component has $T_3 = -\frac{1}{2}$. If the hypercharge line contributes $Y = \frac{1}{2}$, then the section $\eta_0 = v$ satisfies

$$Q\eta_0 = (T_3 + Y)\eta_0 = 0.$$

The IR carrier $\mathbb{CP}^2 \times S_Q^1$ keeps the Q -phase direction and drops the charged weak normal. This gives color times electromagnetism rather than the full electroweak group.

3 KK symmetry as a symmetry of the superstring action

The closed RNS worldsheet action couples directly to the NSNS background,

$$S_{\text{ws}} \supset \frac{1}{4\pi\alpha'} \int_{\Sigma} (G_{MN}(X) + B_{MN}(X)) \partial X^M \bar{\partial} X^N + \text{fermions}. \quad (11)$$

Write local target coordinates as

$$X^M = (x^\mu, y^m), \quad y \in K_6. \quad (12)$$

Let $K_a = K_a^m(y) \partial_m$ be an internal vector field. The transformation

$$\delta y^m = \epsilon^a K_a^m(y) \quad (13)$$

is a symmetry of the sigma model when the background is invariant up to the usual gauge transformations,

$$\mathcal{L}_{K_a} G = 0, \quad \mathcal{L}_{K_a} B = d\Lambda_a, \quad \mathcal{L}_{K_a} \Phi = 0, \quad (14)$$

with the corresponding invariance condition on RR fields in a Green–Schwarz or pure-spinor presentation. This is the worldsheet form of the Kaluza–Klein symmetry.

In target-space language, allowing ϵ^a to depend on x promotes the same symmetry to a gauge redundancy. The metric is written in a connection form,

$$dy^m \longrightarrow dy^m + K_a^m(y) A_\mu^a(x) dx^\mu, \quad (15)$$

so the off-diagonal metric fields A_μ^a are the gauge fields of internal diffeomorphism/isometry symmetry. This is the sense in which the D10 Kaluza–Klein layer is already present in the superstring action: it is the target-space symmetry data of the background on which the worldsheet theory is defined.

For the candidate (1), the closed string sees the base metric, the B -field if present, the dilaton, and the target-space symmetry of $\mathbb{CP}^2 \times \mathbb{CP}^1$. The RR one-form and its Chern data are not ordinary bosonic RNS sigma-model couplings; they enter through the RR sector and through brane/current completions. This distinction matters because the project uses both channels:

$$\text{closed NSNS geometry} \quad + \quad \text{RR/brane/current boundary data.} \quad (16)$$

Proposition 3.1 (Noether symmetry and KK gauge field). *If K_a obeys (14) and preserves the RR background data, then (13) gives a conserved worldsheet current. When the transformation parameter is promoted to a function on M_4 , the metric connection (15) is the four-dimensional gauge field for that symmetry.*

The proof is in Appendix A.

Example 3.2 (The circle model for a KK gauge field). Take one internal circle coordinate $\theta \sim \theta + 2\pi$ and metric

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu + r^2 (d\theta + A_\mu dx^\mu)^2.$$

The local change $\theta \mapsto \theta + \lambda(x)$ is absorbed by

$$A \mapsto A - d\lambda.$$

Thus a coordinate symmetry of the internal circle becomes a $U(1)$ gauge symmetry in the lower-dimensional target. The nonabelian case replaces ∂_θ by the Killing fields K_a on K_6 .

4 The D0 probe

In Type IIA the D0-brane is electrically charged under the RR one-form:

$$S_{\text{D0,WZ}} = \mu_0 \int_{W_1} C_1. \quad (17)$$

Equation (3) identifies C_1 as the connection of the eleventh circle. Therefore D0 charge is M-circle Kaluza–Klein momentum,

$$Q_{\text{D0}} \longleftrightarrow P_\psi. \quad (18)$$

In the ten-dimensional geometry $M_4 \times K_6$, a D0 is a pointlike probe of the RR connection. It measures the same $F_2 = dC_1$ period data appearing in (4),

$$\int_{\Sigma_2} F_2 \in 2\pi\mathbb{Z}. \quad (19)$$

It does not by itself select the weak current sectors. Its role is the electric/M-circle side of the D10–D11 map: as the D0 tower becomes light, the eleventh direction opens.

Proposition 4.1 (D0 charge as M-circle momentum). *For an eleven-dimensional particle or brane state with momentum n/R_{11} around the M-circle, dimensional reduction gives a ten-dimensional state with n units of electric charge under C_1 . The worldline coupling is (17), and the tower mass spacing is $1/R_{11}$.*

The proof is in Appendix A.

Example 4.2 (Aharonov–Bohm test for the RR connection). Let a D0 worldline γ encircle a loop in M_4 on which $\oint_\gamma C_1 = \alpha$. The Wess–Zumino term contributes the phase

$$\exp(i\mu_0\alpha).$$

If a two-chain Σ_2 spans two alternative paths, their relative phase is $\exp(i\mu_0 \int_{\Sigma_2} F_2)$. The D0 therefore measures the M-circle connection, not the Q -boundary line bundle by itself.

5 The space-filling side

The phrase “space-filling brane” has two uses that must be separated. In a ten-dimensional open-string control problem, a D9-brane has Neumann boundary conditions in all target directions,

$$\partial_\sigma X^M|_{\partial\Sigma} = 0, \quad M = 0, \dots, 9. \quad (20)$$

This is the all-Neumann endpoint. A D0 has Neumann boundary condition only along its worldline and Dirichlet boundary conditions along the other spatial directions. In the Type IIA candidate, however, the active spacetime-filling object is not a literal D9. Type IIA carries even D-branes, so the retained object is D8-like:

$$\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1. \quad (21)$$

Its dimension count is

$$\dim_{\mathbb{R}} M_4 + \dim_{\mathbb{R}} \mathbb{CP}^2 + \dim_{\mathbb{R}} S_Q^1 = 4 + 4 + 1 = 9, \quad (22)$$

so it is a D8-type worldvolume. It fills the observed spacetime and wraps the five-dimensional internal carrier

$$K_{5,\text{eff}} = \mathbb{CP}^2 \times S_Q^1, \quad G_{D9} = \frac{SU(3)_c \times U(1)_Q}{\mathbb{Z}_3}. \quad (23)$$

It is codimension one in K_6 ,

$$\mathbb{CP}^2 \times S_Q^1 \subset \mathbb{CP}^2 \times \mathbb{CP}^1. \quad (24)$$

The boundary-condition comparison is

object	Neumann directions	Dirichlet directions
D0	W_1	M_4 spatial + K_6
D8-like	$M_4 + \mathbb{CP}^2 + S_Q^1$	one weak/projective normal
D9 control	$M_4 + K_6$	$\emptyset$.

(25)

Thus the literal space-filling brane is an all-Neumann control, while $\mathcal{W}_9$ is the Type IIA object that can live in the same D10 KK space and still respect the parity of the active theory.

Proposition 5.1 (Boundary-condition and parity check). *The open-string variational principle assigns Neumann boundary conditions along a D-brane worldvolume and Dirichlet boundary conditions in the normal directions. Since Type IIA RR potentials have odd degree in the democratic description, Type IIA D-branes have even spatial dimension. Therefore the object $\mathcal{W}_9$ in (21) is the Type IIA spacetime-filling source in this geometry, while a literal D9 is a Type IIB/type-I control row.*

The proof is in Appendix A.

Example 5.2 (Local coordinates near the D8-like boundary). Choose local coordinates

$$(x^0, \dots, x^3; z^1, \dots, z^4; \phi; n)$$

on $M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1$, where ϕ parameterizes the S^1_Q inside $\mathbb{CP}^1$ and n is the weak/projective normal. The D8-like boundary has Neumann conditions on x^μ, z^i, ϕ and a Dirichlet condition on n . A D9 control would also make n Neumann; a D0 would keep only its worldline Neumann.

6 Wess–Zumino charge and the Q boundary

The D8-like boundary couples to the total RR potential through the Wess–Zumino skeleton

$$S_{\text{WZ}} = \mu_8 \int_{\mathcal{W}_9} C \wedge \text{ch}(E_Q) \sqrt{\frac{\widehat{A}(T\mathcal{W}_9)}{\widehat{A}(N\mathcal{W}_9)}}. \quad (26)$$

The project boundary restricts the neutral line data to

$$c_1(E_Q) = 6h_Q. \quad (27)$$

Since $\mathbb{CP}^2$ is not spin, the boundary gauge field is a spin-c connection. Write the determinant class as

$$x = (2m + 1)h_Q, \quad x \equiv w_2(T\mathbb{CP}^2) \pmod{2}. \quad (28)$$

The WZ curvature is shifted to

$$\frac{F_Q}{2\pi} = 6h_Q + \frac{x}{2}. \quad (29)$$

For the minimal lift $m = 0$,

$$\frac{F_Q}{2\pi} = \frac{13}{2}h_Q, \quad 2\frac{F_Q}{2\pi} = 13h_Q = (P + Q)h_Q. \quad (30)$$

This ties the D8-like boundary to the same $P : Q = 6 : 7$ package as the RR class (4).

The same D8-like object can source a Romans-mass/domain-wall channel:

$$dF_0 = N_8 \delta_{\mathcal{W}_9}. \quad (31)$$

In the Type I' convention used by the project, the local net eight-charge is

$$q_8^{\text{net}} = N_8 - 16. \quad (32)$$

Thus the D8-like boundary is not a second copy of the D0 sector. It is the codimension-one source and boundary-current carrier in the same D10 space.

Proposition 6.1 (Spin-c shift on the $\mathbb{CP}^2$ carrier). *On $\mathbb{CP}^2$, $w_2(T\mathbb{CP}^2) = h \pmod{2}$. A spin-c determinant line therefore has odd first Chern class $x = (2m + 1)h_Q$. Combining this with $c_1(E_Q) = 6h_Q$ gives (29); for $m = 0$ it gives (30).*

The proof is in Appendix A.

Example 6.2 (Half-integral flux on a projective line). Let $\ell \simeq \mathbb{CP}^1$ be a projective line in the $\mathbb{CP}^2$ factor and $\int_\ell h_Q = 1$. The minimal spin-c lift gives

$$\frac{1}{2\pi} \int_\ell F_Q = \frac{13}{2}.$$

The doubled class is integral:

$$2 \frac{1}{2\pi} \int_\ell F_Q = 13.$$

This is the local graduate-level check that the Q -boundary flux is spin-c rather than an ordinary integral line-bundle flux.

7 Comparison

Object	Worldvolume	Coupling	KK role
D0	$W_1 \subset M_4 \times K_6$	$\mu_0 \int C_1$	Electric probe of the M-circle connection; $Q_{D0} = P_\psi$.
D8-like boundary	$M_4 \times \mathbb{CP}^2 \times S_Q^1$	S_{WZ} in Eq. (26)	Codimension-one source, Q -boundary carrier, and host for Chan–Paton/current sectors.
D9 control	$M_4 \times K_6$	All-Neumann open-string control	Type IIB/type-I space-filling endpoint, not the active Type IIA source.

All three rows refer to the same target space $M_4 \times K_6$, but they probe different data. The D0 tests the electric RR one-form and the M-circle period. The D8-like boundary wraps the D9 endpoint carrier and can support the open string endpoint/current data. The D9 row is useful for boundary-condition bookkeeping but is not the Type IIA brane used in the current candidate.

Proposition 7.1 (Separation of the three rows). *The D0, D8-like boundary, and D9 control row are distinguished by three invariants: worldvolume dimension, RR coupling, and boundary condition in the weak/projective normal. These invariants separate the electric M-circle probe, the codimension-one Q -boundary source, and the all-Neumann control endpoint.*

The proof is in Appendix A.

Example 7.2 (Three observables in one target space). In $M_4 \times K_6$, a D0 detects $\int_{\Sigma_2} F_2$ through its worldline phase. A D8-like boundary detects the jump of F_0 across the normal direction through (31). A D9 control checks the all-Neumann endpoint condition (20). The observables live in one target space but test different structures.

8 Consequence for the electroweak interface

The Kaluza–Klein symmetry statement gives the correct place for electroweak labels: they must be target-space symmetry, current, or boundary-sector labels, not scalar harmonics on $\mathbb{CP}^1$ alone. The

D0 sector supports the M-circle/RR period side. The D8-like boundary is the natural candidate for the $j = 0, \frac{1}{2}, 1$ action-placement gate because it can carry line-bundle zero-mode spaces

$$V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)), \quad j = 0, \frac{1}{2}, 1, \quad (33)$$

and it can host group-valued endpoint variables whose quantization gives Chan–Paton representations and Wilson-loop insertions $T_a, T_a T_b$.

The derived representation statement is

$$T^a T_a|_{V_j} = j(j+1) \mathbf{1}_{V_j}. \quad (34)$$

This note uses (34) only as representation placement on the D8-like boundary. It does not assume a mass law from this Casimir.

Proposition 8.1 (Borel–Weil placement). *For $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, the vector space $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$ is the irreducible $SU(2)$ representation of spin j . Its dimension is $2j+1$, and the quadratic Casimir acts as (34). Therefore the $j = 0, \frac{1}{2}, 1$ labels can be placed on the D8-like endpoint/current sector without treating them as scalar KK masses.*

The proof is in Appendix A.

Example 8.2 (The first three $\mathbb{CP}^1$ zero-mode spaces). Let $[u : v]$ be homogeneous coordinates on $\mathbb{CP}^1$. Then

j	$H^0(\mathbb{CP}^1, \mathcal{O}(2j))$	dim	$T^a T_a$
0	$\langle 1 \rangle$	1	0
$\frac{1}{2}$	$\langle u, v \rangle$	2	$\frac{3}{4}$
1	$\langle u^2, uv, v^2 \rangle$	3	2.

This is the representation-theoretic location of the project labels. The Casimir values are outputs of the endpoint Hilbert spaces, not input scalar Laplacian eigenvalues.

A Proofs and checks

Example A.1 (How the appendix is used). The main text states each structural claim as a proposition and gives one worked example in the same section. The appendix proves the propositions in the order in which they occur, so every non-tabular assertion used by the article has a local check here.

Proof of Proposition 1.1. On an overlap of two circle-bundle patches, let $\psi_i = \psi_j - \lambda_{ij}$ and $C_i = C_j + d\lambda_{ij}$. Then

$$d\psi_i + C_i = d\psi_j + C_j,$$

so the last term in (3) is globally defined. Its curvature is $dC_i = dC_j$, hence it descends to a global two-form F_2 on K_6 . By Chern–Weil theory, $F_2/2\pi$ represents the first Chern class of the circle bundle when the periods are integral. Reducing the eleven-dimensional metric along the circle gives the ten-dimensional metric, dilaton, and RR one-form C_1 . If an internal vector field is Killing and preserves the cohomology class of F_2 , its flow maps the full reduced background to an equivalent one. These vector fields close under Lie bracket, giving the Kaluza–Klein gauge algebra. $\square$

Proof of Proposition 2.1. For a field on a circle of radius R ,

$$\Phi(x, \theta) = \sum_{n \in \mathbb{Z}} \Phi_n(x) e^{in\theta}$$

and the internal Laplacian contributes n^2/R^2 to the lower-dimensional mass squared. If $E \ll 1/R$, the $n \neq 0$ modes cannot be produced as propagating modes, so the effective description keeps the zero mode. Applying this first to the weak/projective normal leaves $\mathbb{CP}^2 \times S_Q^1$ and the group in (7). Applying it to the M-circle gives the threshold $1/R_{11}$. Above that threshold the D0 momentum tower is part of the propagating spectrum; with $R_{11} = g_s \ell_s$, this is the Witten strong-coupling restoration of the eleventh dimension. $\square$

Proof of Proposition 3.1. Vary the NSNS part of the sigma-model action by $\delta X^M = \epsilon K^M$, where K is internal. The metric variation is $\epsilon(\mathcal{L}_K G)_{MN} \partial X^M \bar{\partial} X^N$, which vanishes when $\mathcal{L}_K G = 0$. The B -field variation is $\epsilon d\Lambda$, whose worldsheet pullback is a total derivative for constant ϵ ; the dilaton is invariant by assumption. Noether's theorem then gives a conserved current. If $\epsilon = \epsilon(x)$, derivatives of ϵ appear. The connection term in (15) cancels those terms under the usual gauge transformation of A_μ^a . Thus the lower-dimensional field A_μ^a is the gauge field for the internal symmetry generated by K_a . $\square$

Proof of Proposition 4.1. For a particle moving in the eleven-dimensional circle metric, the covariant velocity along the circle is $\dot{\psi} + C_\mu \dot{x}^\mu$. The canonical momentum conjugate to ψ is conserved because ψ is a circle coordinate. Quantization on the circle gives $P_\psi = n/R_{11}$. In the first-order worldline action, the term linear in the four-dimensional velocity is $n \int C_1$, which is the electric coupling of a charge- n D0 state. The same quantized momentum gives the tower spacing $1/R_{11}$. $\square$

Proof of Proposition 5.1. The boundary variation of the open-string Polyakov action contains

$$\delta S_{\partial\Sigma} = \frac{1}{2\pi\alpha'} \int_{\partial\Sigma} G_{MN} \partial_\sigma X^N \delta X^M.$$

It vanishes either by imposing $\partial_\sigma X^M = 0$ in a Neumann direction or by fixing $\delta X^M = 0$ in a Dirichlet direction. A Dp -brane has $p+1$ Neumann coordinates. In the democratic Type IIA RR description the potentials have odd degree C_1, C_3, C_5, C_7, C_9 , so a Dp -brane coupling $\int C_{p+1}$ has p even. The worldvolume in (21) is nine-dimensional, hence $p = 8$. A D9 would need ten Neumann coordinates and is the all-Neumann control row rather than the active Type IIA source. $\square$

Proof of Proposition 6.1. The Euler sequence for $\mathbb{CP}^2$ gives

$$c(T\mathbb{CP}^2) = (1+h)^3,$$

after truncating above degree four. Hence $c_1(T\mathbb{CP}^2) = 3h$, and for a complex manifold $w_2 = c_1 \pmod{2}$, so $w_2(T\mathbb{CP}^2) = h \pmod{2}$. A spin- c determinant line must reduce to this class mod 2, so $x = (2m+1)h_Q$. The Freed–Witten shifted curvature is the ordinary integral line-bundle class plus $x/2$, giving

$$\frac{F_Q}{2\pi} = 6h_Q + \frac{(2m+1)h_Q}{2}.$$

At $m = 0$ this is $13h_Q/2$, and doubling gives $13h_Q = (P+Q)h_Q$ for $P : Q = 6 : 7$. $\square$

Proof of Proposition 7.1. The D0 row has one-dimensional worldvolume and couples electrically to C_1 , so it sources the equation dual to $F_2 = dC_1$ and measures the M-circle connection. The D8-like row has nine-dimensional worldvolume, wraps the Q -carrier, and appears in the Bianchi identity $dF_0 = N_8 \delta \mathcal{W}_9$; it is therefore a codimension-one source. The D9 control row has ten Neumann directions and no Type IIA C_{10} coupling in the active setup. Dimension, RR coupling, and normal boundary condition agree with the three entries of the table and separate the rows. $\square$

Proof of Proposition 8.1. Sections of $\mathcal{O}(n)$ over $\mathbb{CP}^1$ are homogeneous polynomials of degree n in two variables u, v . Therefore

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \simeq \text{Sym}^n(\mathbb{C}^2)^*$$

and has dimension $n + 1$. The natural $SU(2)$ action on (u, v) induces the irreducible symmetric-power representation of highest weight n . For $n = 2j$, this is the spin- j representation. With the usual normalization of the $SU(2)$ generators, the quadratic Casimir on spin j is $j(j + 1)\mathbf{1}$, proving (34). Quantized endpoint variables valued in the same representation insert the matrices T_a and $T_a T_b$ on Chan–Paton/Wilson factors. $\square$

Part II

A D10 boundary action on $M_4 \times \mathbb{CP}^2 \times S_Q^1$

Abstract

This note constructs the boundary action that was used only schematically in the D10 Kaluza–Klein article. The boundary is

$$\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1 \subset M_4 \times \mathbb{CP}^2 \times \mathbb{CP}^1.$$

The action contains the D8-like DBI/Wess–Zumino source, a spin-c Q -line bundle, a group-valued endpoint degree of freedom whose quantization gives $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$, and a quadratic boundary/current kernel built from the same zero-mode operator that appears in the Wilson expansion. The construction preserves the $Q = T_3 + Y$ photon boundary and gives a finite moduli space: one $U(1)_Q$ holonomy, discrete spin-c and D8-charge data, the common boundary scale, a direct electroweak-vacuum modulus, a scalar-curvature modulus for the Higgs mode, and a current-metric deformation whose canonical slice is the unit zero-mode map.

1 Boundary data

Let

$$X_{10} = M_4 \times K_6, \quad K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1,$$

and let the D8-like boundary worldvolume be

$$\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1. \quad (35)$$

The circle S_Q^1 is the diagonal electromagnetic circle inside the weak $\mathbb{CP}^1$ factor after choosing the neutral section

$$Q = T_3 + Y, \quad Q\eta_0 = 0.$$

The spin-c Q -line on the $\mathbb{CP}^2$ carrier is denoted $E_Q \rightarrow \mathcal{W}_9$. Its curvature is

$$\frac{F_Q}{2\pi} = 6h_Q + \frac{x}{2}, \quad x = (2m+1)h_Q, \quad m \in \mathbb{Z}. \quad (36)$$

The endpoint Hilbert spaces are the Borel–Weil spaces

$$V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)), \quad \mathcal{J}_2 = \left\{0, \frac{1}{2}, 1\right\}. \quad (37)$$

The boundary Chan–Paton bundle used below is

$$\mathcal{V}_\partial = E_Q \otimes \bigoplus_{j \in \mathcal{J}_2} V_j. \quad (38)$$

Proposition 1.1 (Endpoint sectors). *For $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, $V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j))$ is the spin- j irreducible representation of $SU(2)$. The subset $\mathcal{J}_2$ is the level-two finite endpoint set, and on V_j*

$$\sum_{a=1}^3 T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1}_{V_j}.$$

Proof. Sections of $\mathcal{O}(n)$ on $\mathbb{CP}^1$ are homogeneous polynomials of degree n in two variables. Thus

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) \simeq \text{Sym}^n(\mathbb{C}^2)^*,$$

which is the irreducible $SU(2)$ representation of highest weight n . Putting $n = 2j$ gives spin j and dimension $2j + 1$. The standard normalization of the $SU(2)$ generators gives the Casimir (1.1). The level-two current selector keeps $0 \leq j \leq 1$. $\square$

Example 1.2 (The three retained spaces). With homogeneous coordinates $[u : v]$ on $\mathbb{CP}^1$,

j	V_j	$\dim V_j$	$j(j+1)$
0	$\langle 1 \rangle$	1	0
$\frac{1}{2}$	$\langle u, v \rangle$	2	$\frac{3}{4}$
1	$\langle u^2, uv, v^2 \rangle$	3	2.

These are the photon, fundamental endpoint, and adjoint-bilinear slots used by the boundary action.

2 Endpoint action

Attach to each boundary component of the open string a group-valued variable

$$g_j(\tau) \in SU(2)$$

and choose the coadjoint weight

$$\lambda_j = jH, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} / 2.$$

Let $\mathcal{A}_\tau = A_\tau^a T_a^{(j)}$ be the pullback to $\partial\Sigma$ of the weak/current connection on $\mathcal{W}_9$. The endpoint action is

$$S_{\text{end}}^{(j)} = \int_{\partial\Sigma} d\tau \left\langle \lambda_j, -i g_j^{-1} (\partial_\tau + i \mathcal{A}_\tau) g_j \right\rangle. \quad (39)$$

For a second endpoint the sign is reversed, giving the dual representation in the oriented open-string sector.

Proposition 2.1 (Chan–Paton and Wilson factors). *Quantization of (39) gives the Hilbert space V_j . The boundary path integral with background $\mathcal{A}_\tau$ gives the Wilson operator*

$$W_j[\mathcal{A}] = \text{Tr}_{V_j} P_{\partial\Sigma} \exp \left(i \int_{\partial\Sigma} d\tau A_\tau^a T_a^{(j)} \right). \quad (40)$$

Its first and second functional variations insert $T_a^{(j)}$ and the ordered product $T_a^{(j)} T_b^{(j)}$.

Proof. The first term in (39) is the prequantum action on the coadjoint orbit through λ_j . Integrality of the symplectic class is $2j \in \mathbb{Z}_{\geq 0}$, and geometric quantization gives the Borel–Weil space $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$. Coupling to $\mathcal{A}_\tau$ replaces ∂_τ by a covariant derivative, so time evolution along the boundary is the ordered exponential in (40). Differentiating this ordered exponential with respect to A_τ^a inserts the corresponding representation generator at the differentiation point. Differentiating twice inserts the ordered pair. $\square$

Example 2.2 (Generator insertions). Set $A_\tau^a = \mu \xi^a(\tau)$. Then

$$\left. \frac{\delta W_j}{\delta \xi^a(\tau)} \right|_{\xi=0} = i\mu \text{Tr}_{V_j}(T_a^{(j)})$$

with the trace replaced by the ordered insertion in amplitudes containing other boundary operators. The second variation gives

$$-\mu^2 \text{Tr}_{V_j} \left(P_{\partial\Sigma} T_a^{(j)} T_b^{(j)} \right).$$

These are the Chan–Paton tensors used by the current kernel below.

3 The Q -preserving boundary field

The boundary-changing field selecting the electromagnetic endpoint is a section

$$\eta \in H^0(\mathbb{CP}^1, \mathcal{O}(1)) \otimes L_Y,$$

where Y acts as $+\frac{1}{2}$ on L_Y . Choose

$$\eta_0 = v, \quad T_3 \eta_0 = -\frac{1}{2} \eta_0, \quad Y \eta_0 = \frac{1}{2} \eta_0.$$

Then (1) holds. The boundary term on $\mathcal{W}_9$ is

$$S_\eta = \int_{\mathcal{W}_9} d^9 \xi \sqrt{|g_9|} (\langle D\eta, D\eta \rangle + V_\eta(\langle \eta, \eta \rangle)), \quad (41)$$

where

$$D = d + iA^a T_a + iB Y, \quad V_\eta(r) = \lambda_\eta (r - v_Q^2)^2, \quad \lambda_\eta > 0.$$

On the Q -boundary slice the unbroken connection is the component along

$$A_Q Q, \quad Q = T_3 + Y.$$

Proposition 3.1 (Photon protection). *On the boundary slice $\eta = \eta_0$, the term $\langle D\eta, D\eta \rangle$ contains no mass term for A_Q . The orthogonal weak-hypercharge combination acts nontrivially on η_0 .*

Proof. The part of $D\eta_0$ proportional to the Q -connection is

$$iA_Q Q \eta_0.$$

Since $Q\eta_0 = 0$, this contribution vanishes. Any generator R whose action on η_0 is nonzero contributes $\langle R\eta_0, R\eta_0 \rangle A_R^2$, so the corresponding connection is not protected by the neutral boundary condition. $\square$

Example 3.2 (The doublet). In the basis (u, v) of $H^0(\mathbb{CP}^1, \mathcal{O}(1))$, T_3 has eigenvalues $+\frac{1}{2}$ and $-\frac{1}{2}$. Tensoring with a hypercharge line of $Y = \frac{1}{2}$, the lower component v has $Q = 0$. The upper component has $Q = 1$ and is not part of the neutral boundary slice.

4 Current map and quadratic kernel

Let e_a be an orthonormal frame for the retained weak/current image directions. For each $j \in \mathcal{J}_2$, define

$$A_j : V_j \longrightarrow V_j \otimes \mathbb{R}^3, \quad A_j t = \sum_{a=1}^3 T_a^{(j)} t \otimes e_a. \quad (42)$$

Then

$$A_j^\dagger A_j = J_j \mathbf{1}_{V_j}, \quad J_j = j(j+1). \quad (43)$$

For $j > 0$, let $\text{Im } A_j$ be the image subbundle and let

$$P_j = \frac{1}{J_j} A_j A_j^\dagger$$

be the image projector. For $j = 0$, $A_0 = 0$.

Let p_j be the open boundary/vector channel with values in V_j , and let q_j be the image channel with values in $\text{Im } A_j$. Let B be the positive four-dimensional inverse propagator variable restricted to the boundary mode. The quadratic boundary action is

$$S_{\text{quad}}^{(j)} = \int_{\mathcal{W}_9} d^9 \xi \sqrt{|g_9|} [\langle p_j, B p_j \rangle + \langle q_j, (B + \mu^2 J_j) q_j \rangle - \mu^2 \langle q_j, A_j p_j \rangle - \mu^2 \langle A_j p_j, q_j \rangle]. \quad (44)$$

The coefficient μ is the common boundary scale.

Proposition 4.1 (Feshbach reduction). *Eliminating q_j from (44) gives the p_j -channel inverse propagator*

$$\Gamma_j(B) = B - \mu^4 A_j^\dagger (B + \mu^2 J_j)^{-1} A_j. \quad (45)$$

On V_j , this is

$$\Gamma_j(B) = B - \frac{\mu^4 J_j}{B + \mu^2 J_j}. \quad (46)$$

With $X = B/\mu^2$, the pole equation is

$$X = \frac{J_j}{X + J_j}. \quad (47)$$

For $j = 0$, the coupling vanishes and the boundary photon slot remains massless.

Proof. The Euler equation for q_j from (44) is

$$(B + \mu^2 J_j) q_j = \mu^2 A_j p_j.$$

Substituting

$$q_j = \mu^2 (B + \mu^2 J_j)^{-1} A_j p_j$$

back into the quadratic form gives (45). Since B is scalar on the retained finite endpoint space and $A_j^\dagger A_j = J_j \mathbf{1}$, (46) follows. Setting $\Gamma_j(B) = 0$ and dividing by μ^2 gives (47). For $j = 0$, $J_0 = 0$ and $A_0 = 0$. $\square$

Example 4.2 (The $j = \frac{1}{2}$ slot). For $j = \frac{1}{2}$, $J = 3/4$. The pole equation is

$$X = \frac{3/4}{X + 3/4}, \quad X^2 + \frac{3}{4}X - \frac{3}{4} = 0.$$

This comes from the single map

$$A_{1/2}t = \sum_{a=1}^3 \frac{\sigma_a}{2} t \otimes e_a, \quad A_{1/2}^\dagger A_{1/2} = \frac{3}{4} \mathbf{1}_2.$$

5 The full $\mathcal{W}_9$ action

The constructed D10 boundary action is

$$S_{\mathcal{W}_9} = N_8(S_{\text{DBI}} + S_{\text{WZ}}) + S_\eta + \sum_{j \in \mathcal{J}_2} \left(S_{\text{end}}^{(j)} + S_{\text{quad}}^{(j)} \right). \quad (48)$$

The D8-like source terms are

$$S_{\text{DBI}} = -T_8 \int_{\mathcal{W}_9} e^{-\Phi} \sqrt{-\det(\iota^*(G + B) + 2\pi\alpha' F_Q)} \quad (49)$$

and

$$S_{\text{WZ}} = \mu_8 \int_{\mathcal{W}_9} C \wedge \text{ch}(E_Q) \sqrt{\frac{\widehat{A}(T\mathcal{W}_9)}{\widehat{A}(N\mathcal{W}_9)}}. \quad (50)$$

The local Romans source convention is

$$dF_0 = N_8 \delta_{\mathcal{W}_9}. \quad (51)$$

Proposition 5.1 (Boundary-action package). *The action (48) has the four required properties:*

- (i) *it is supported on $\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1$,*
- (ii) *it carries the endpoint spaces $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$,*
- (iii) *it preserves the Q photon boundary,*
- (iv) *it produces the Chan–Paton/Wilson insertions $T_a, T_a T_b$.*

The same insertions define the current map A_j in the quadratic kernel.

Proof. The support is $\mathcal{W}_9$ by (35), (49), and (50). The integer N_8 multiplies the source and appears in (51). The endpoint Hilbert spaces are V_j by Proposition 1.1 and Proposition 2.1. The Q boundary is protected by Proposition 3.1. The Wilson insertions are the functional variations in Proposition 2.1. Contracting the first variation with the orthonormal current frame gives A_j , and the second variation gives $A_j A_j^\dagger$, which is the tensor in (44). $\square$

Example 5.2 (Worksheet endpoint on the D8-like boundary). For a boundary path $X(\tau) \in \mathcal{W}_9$, the full boundary factor in a fixed j -sector is

$$\exp \left(i S_{\text{end}}^{(j)}[g_j, X] \right) \exp \left(i S_{\text{quad}}^{(j)}[p_j, q_j] \right).$$

After quantizing g_j , the first exponential becomes the Wilson operator $\text{Tr}_{V_j} P \exp(i \int A_\tau^a T_a^{(j)} d\tau)$. The quadratic term then uses the same $T_a^{(j)}$ in A_j .

6 Moduli space

The boundary data split into discrete topological data and continuous flat or scale data.

Topological and representation data

The internal boundary carrier is $K_5 = \mathbb{CP}^2 \times S_Q^1$. Since

$$H^1(K_5, \mathbb{Z}) \simeq \mathbb{Z}, \quad H^2(K_5, \mathbb{Z}) \simeq \mathbb{Z},$$

a $U(1)_Q$ line bundle has one Chern-class integer on $\mathbb{CP}^2$ and one flat holonomy on S_Q^1 . The spin-c shift fixes the Chern class to the discrete family (36). The endpoint representation data are also discrete:

$$j \in \mathcal{J}_2, \quad 2j = 0, 1, 2.$$

For each j , $\text{Pic}^{2j}(\mathbb{CP}^1)$ is a point, so the bundle $\mathcal{O}(2j)$ has no continuous line-bundle modulus on $\mathbb{CP}^1$.

Continuous data

The continuous moduli in the minimal boundary action are:

$$\theta_Q = \oint_{S_Q^1} A_Q \in \mathbb{R}/2\pi\mathbb{Z}, \quad \mu \in \mathbb{R}_{>0}, \quad v_Q \in \mathbb{R}_{>0}, \quad \lambda_H \in \mathbb{R}_{>0}.$$

Here v_Q is the norm of the neutral boundary section and λ_H denotes the radial scalar curvature after canonical normalization. In the one-field limit,

$$v_{\text{EW}} = \sqrt{2} v_Q, \quad V_H(h) = \frac{\lambda_H}{4} (h^2 - v_{\text{EW}}^2)^2, \quad M_H^2 = 2\lambda_H v_{\text{EW}}^2.$$

If one keeps the orthogonal auxiliary complement to $V_j \otimes \mathbb{R}^3$, there is also a positive gap $M_\perp \in \mathbb{R}_{>0}$. If one allows a noncanonical current metric

$$\langle e_a, e_b \rangle_{\text{cur}} = \kappa_{\text{cur}} \delta_{ab},$$

then the current map deforms to

$$A_j^\dagger A_j = \kappa_{\text{cur}} J_j \mathbf{1}_{V_j}.$$

The one-operator action keeps the Regge-compatible relation between off-diagonal and image-channel terms; with $c = \sqrt{\kappa_{\text{cur}}}$, the pole equation is

$$X^2 + c^2 J_j X - c^2 J_j = 0. \quad (52)$$

The action (48) is the canonical slice

$$\kappa_{\text{cur}} = 1.$$

Proposition 6.1 (Minimal moduli space). *After quotienting by the $SU(2) \times U(1)_Y$ gauge action that fixes the neutral boundary direction, the minimal moduli space of the boundary action is*

$$\mathcal{M}_\partial^{\min} = \bigsqcup_{m \in \mathbb{Z}, N_8 \in \mathbb{Z}} (U(1)_{\theta_Q} \times \mathbb{R}_{>0, \mu} \times \mathbb{R}_{>0, v_Q} \times \mathbb{R}_{>0, \lambda_H}), \quad (53)$$

with the finite endpoint set $\mathcal{J}_2 = \{0, \frac{1}{2}, 1\}$ fixed. The enlarged metric-deformation space is

$$\widetilde{\mathcal{M}}_\partial = \mathcal{M}_\partial^{\min} \times \mathbb{R}_{>0, \kappa_{\text{cur}}} \quad \text{or} \quad \mathcal{M}_\partial^{\min} \times \mathbb{R}_{>0, \kappa_{\text{cur}}} \times \mathbb{R}_{>0, M_\perp} \quad (54)$$

when the current metric or auxiliary gap is not fixed.

Proof. The Kunneth formula gives the cohomology groups above for $\mathbb{CP}^2 \times S^1$. A fixed Chern class on $\mathbb{CP}^2$ leaves flat $U(1)$ -connections classified by $\text{Hom}(H_1(S_Q^1), U(1)) \simeq U(1)$, giving θ_Q . The spin-c lift integer m and the D8 charge N_8 are discrete. Borel–Weil and the level-two restriction make the endpoint set finite, and $\text{Pic}^{2j}(\mathbb{CP}^1)$ is a point. The neutral vector η_0 has stabilizer $U(1)_Q$; the orbit $(SU(2) \times U(1)_Y)/U(1)_Q$ is gauge, so it is removed in the gauge quotient. The action leaves the positive scale μ , the neutral amplitude v_Q , and the radial curvature λ_H . The optional deformations κ_{cur} and $M_\perp$ are positive real parameters by definition. $\square$

Example 6.2 (Flat Q holonomy). For a state of Q -charge q , the flat connection on S_Q^1 contributes

$$\exp(iq\theta_Q).$$

The selected neutral boundary field has $q = 0$, so this holonomy does not give the photon a mass. Charged endpoint states acquire the usual Wilson phase.

7 Compatibility with the de Vries data

For the current-metric deformation $c = \sqrt{\kappa_{\text{cur}}}$, define

$$X_\pm(J; c) = \frac{-c^2 J \pm \sqrt{c^4 J^2 + 4c^2 J}}{2}. \quad (55)$$

The positive pole branch of the boundary kernel is $X_+(J; c)$. The signed two-channel Hessian uses the pair $X_\pm(J; c)$.

Proposition 7.1 (de Vries-compatible slice). *The boundary moduli are compatible with the de Vries electroweak data on the slice*

$$\kappa_{\text{cur}} = 1, \quad \frac{M_W^2}{\mu^2} = X_+\left(\frac{3}{4}; 1\right), \quad \frac{M_Z^2}{\mu^2} = X_+(2; 1). \quad (56)$$

On the direct boundary-scalar enhancement, the electroweak vacuum and Higgs curvature are placed on the negative branch by

$$\frac{v_{\text{EW}}^2}{2\mu^2} = |X_-(2; 1)|, \quad \frac{M_H^2}{\mu^2} = \left|X_-\left(\frac{3}{4}; 1\right)\right|, \quad v_{\text{EW}} = \sqrt{2} v_Q. \quad (57)$$

Equivalently, the radial quartic modulus is

$$\lambda_H = \frac{|X_-(\frac{3}{4}; 1)|}{4|X_-(2; 1)|}. \quad (58)$$

Proof. The W and Z equations are Proposition 4.1 evaluated at $J = \frac{3}{4}$ and $J = 2$ with the canonical current metric $\kappa_{\text{cur}} = 1$. The weak-angle ratio is then

$$\sin^2 \theta_W = 1 - \frac{X_+(\frac{3}{4}; 1)}{X_+(2; 1)}.$$

The same signed quadratic operator has the negative eigenvalues $X_-(J; 1)$. Taking the $J = 2$ negative branch as the direct boundary order-parameter scale gives $v_{\text{EW}}^2/2 = \mu^2 |X_-(2; 1)|$. Taking the $J = \frac{3}{4}$ negative branch as the radial scalar scale gives the second equation in (57). Combining this with the canonical quartic relation $M_H^2 = 2\lambda_H v_{\text{EW}}^2$ gives (58). $\square$

Example 7.2 (Numerical compatibility). At $\kappa_{\text{cur}} = 1$,

$$X_+\left(\frac{3}{4}; 1\right) = \frac{\sqrt{57}-3}{8}, \quad X_+(2; 1) = \sqrt{3} - 1,$$

so

$$\sin^2 \theta_W = 0.2231013223 \dots$$

The negative-branch assignments give

$$|X_-(2; 1)| = 1 + \sqrt{3}, \quad \left|X_-\left(\frac{3}{4}; 1\right)\right| = \frac{\sqrt{57}+3}{8},$$

and hence

$$\lambda_H = 0.12067 \dots$$

If μ is fixed from the Z pole, the $J = 2$ negative branch gives the electroweak order-parameter scale $v_{\text{EW}}/\sqrt{2}$, while the $J = \frac{3}{4}$ negative branch gives a Higgs-scale radial mode. This is the direct boundary-scalar reading. A Yukawa reading would replace (57) by a fermion-coupling condition and is not part of this boundary action.

8 Summary of the construction

The action is the D8-like source action (49)–(50), the neutral Q -boundary term (41), the endpoint coadjoint-orbit action (39), and the quadratic current kernel (44). The endpoint action quantizes to $H^0(\mathbb{CP}^1, \mathcal{O}(2j))$. Its Wilson expansion produces the T_a and $T_a T_b$ insertions. The same generators define

$$A_j = \sum_a T_a^{(j)} \otimes e_a,$$

and the Feshbach reduction gives the pole equation

$$X = \frac{j(j+1)}{X + j(j+1)}$$

on the canonical current-metric slice. The $j = 0$ sector is massless, and the independent Q -boundary proof shows that the massless state is the photon boundary, not an accidental zero of the auxiliary kernel. The enhanced moduli space adds v_Q and λ_H , so the same boundary package can place the de Vries weak angle, the electroweak vacuum, and the Higgs-scale radial mode on one compatibility slice.

Proposition 8.1 (Construction audit). *The action (48), together with the compatibility slice (56)–(57), satisfies the article’s four construction requirements and its scalar enhancement:*

- (i) $\mathcal{W}_9 = M_4 \times \mathbb{CP}^2 \times S_Q^1$,
- (ii) $V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j))$, $j = 0, \frac{1}{2}, 1$,
- (iii) $Q\eta_0 = 0$,
- (iv) $W_j[\mathcal{A}]$ inserts T_a and $T_a T_b$,
- (v) $\kappa_{\text{cur}} = 1$, v_Q , λ_H give the de Vries-compatible weak, vacuum, and Higgs data.

Proof. Item (i) is the support statement in (35) and Proposition 5.1. Item (ii) is Proposition 1.1. Item (iii) is Proposition 3.1. Item (iv) is Proposition 2.1. Item (v) is Proposition 7.1. Thus each summary claim is already proved by a proposition in the main text. $\square$

Example 8.2 (Canonical compatibility checklist). At the canonical point, choose

$$j \in \left\{0, \frac{1}{2}, 1\right\}, \quad \kappa_{\text{cur}} = 1, \quad Q\eta_0 = 0.$$

Then $j = 0$ gives the photon slot, $j = \frac{1}{2}$ gives

$$X_+\left(\frac{3}{4}; 1\right) = \frac{\sqrt{57} - 3}{8},$$

and $j = 1$ gives

$$X_+(2; 1) = \sqrt{3} - 1.$$

The same point keeps $v_{\text{EW}} = \sqrt{2}v_Q$ and $\lambda_H = |X_-(\frac{3}{4}; 1)|/(4|X_-(2; 1)|)$ as scalar-sector entries rather than inserting them into the Wilson operator.

Part III

Current metric and coupled vacuum potential

1 Canonical current metric

Let K_a , $a = 1, 2, 3$, be the weak Killing/current generators on the internal current space, with the same metric that appears in the Kaluza–Klein reduction. Define

$$G_{ab}^{(2)} = \frac{1}{\text{Vol}(\widehat{K}_6)} \int_{\widehat{K}_6} \sqrt{g} g_{mn} K_a^m K_b^n. \quad (59)$$

On the $SU(2)$ -symmetric weak line,

$$G_{ab}^{(2)} = L_2^2 \delta_{ab}. \quad (60)$$

The orthonormal frame is

$$e_a = \frac{K_a}{L_2}, \quad \langle e_a, e_b \rangle = \delta_{ab}. \quad (61)$$

For $j \in \{0, \frac{1}{2}, 1\}$, set

$$A_j : V_j \longrightarrow V_j \otimes \mathbb{R}^3, \quad A_j t = \sum_{a=1}^3 T_a^{(j)} t \otimes e_a. \quad (62)$$

Proposition 1.1 (Canonical current normalization). *With (61),*

$$A_j^\dagger A_j = j(j+1) \mathbf{1}_{V_j}. \quad (63)$$

Consequently the current-metric factor in the boundary kernel is

$$\kappa_{\text{cur}} = 1. \quad (64)$$

A value $\kappa_{\text{cur}} \neq 1$ is an added boundary current metric rather than the Kaluza–Klein current metric.

Proof. Using the orthonormal frame,

$$A_j^\dagger A_j = \sum_{a,b} T_a^{(j)} T_b^{(j)} \langle e_a, e_b \rangle = \sum_a T_a^{(j)} T_a^{(j)}.$$

The last operator is the $SU(2)$ quadratic Casimir on the spin- j representation, hence $j(j+1) \mathbf{1}_{V_j}$. A multiplicative factor κ_{cur} appears only after replacing the image metric by $\langle e_a, e_b \rangle_{\text{cur}} = \kappa_{\text{cur}} \delta_{ab}$, which is not (59). $\square$

Example 1.2 (Round $\mathbb{CP}^1$ control). For a round $\mathbb{CP}^1 \simeq S^2$ of radius R_w ,

$$\frac{1}{\text{Vol}(\mathbb{CP}^1)} \int_{\mathbb{CP}^1} \sqrt{g} g(K_a, K_b) = \frac{2}{3} R_w^2 \delta_{ab}.$$

Thus $L_2^2 = 2R_w^2/3$, and $e_a = K_a/L_2$ gives an orthonormal current frame. If

$$L_{2,\text{BL}}^2 = L_{2,\text{horiz}}^2 + L_{2,\text{vert}}^2,$$

then the total length changes the gauge coupling, while the normalized Casimir in (63) is unchanged.

2 Coupled vacuum potential

Couple the boundary scale to the closed-string moduli by writing it as $\mu_B(\rho, \tau)$. The Type IIA bulk scaling potential is

$$V_{\text{bulk}} = -B_R \tau^{-2} \rho^{-2} + B_H \tau^{-2} \rho^{-6} + B_0 \tau^{-4} \rho^6 + B_2 \tau^{-4} \rho^2 \\ + B_4 \tau^{-4} \rho^{-2} + B_6 \tau^{-4} \rho^{-6} + B_{\text{loc}} \tau^{-3}. \quad (65)$$

Set

$$J_v = 1 + \sqrt{3}, \quad J_H = \frac{3 + \sqrt{57}}{8}, \quad \lambda_{\text{dV}} = \frac{J_H}{4J_v}. \quad (66)$$

A selector potential compatible with the boundary construction is

$$V_{\text{sel}} = C_{\text{cur}}(\rho, \tau)(\kappa_{\text{cur}} - 1)^2 \\ + C_v(\rho, \tau)(v_Q^2 - \mu_B(\rho, \tau)^2 J_v)^2 \\ + C_H(\rho, \tau)(\lambda_H - \lambda_{\text{dV}})^2, \quad (67)$$

with $C_{\text{cur}}, C_v, C_H > 0$. The radial Higgs potential is

$$V_{\text{rad}}(h) = \frac{\lambda_H}{4}(h^2 - 2v_Q^2)^2, \quad v_{\text{EW}} = \sqrt{2} v_Q. \quad (68)$$

Use

$$V_{\text{eff}} = V_{\text{bulk}} + V_{\text{sel}} + V_{\text{rad}}. \quad (69)$$

Proposition 2.1 (Vacuum selector). *At the radial minimum $h^2 = 2v_Q^2$, the nonzero electroweak branch satisfies*

$$\kappa_{\text{cur}} = 1, \quad v_Q^2 = \mu_B(\rho, \tau)^2(1 + \sqrt{3}), \quad \lambda_H = \frac{3 + \sqrt{57}}{32(1 + \sqrt{3})}. \quad (70)$$

It follows that

$$v_{\text{EW}} = \sqrt{2} v_Q, \quad M_H^2 = 2\lambda_H v_{\text{EW}}^2. \quad (71)$$

Proof. At $h^2 = 2v_Q^2$, the first derivative of V_{rad} in the radial direction vanishes. The selector derivatives are

$$\partial_{\kappa_{\text{cur}}} V_{\text{eff}} = 2C_{\text{cur}}(\rho, \tau)(\kappa_{\text{cur}} - 1), \\ \partial_{\lambda_H} V_{\text{eff}} = 2C_H(\rho, \tau)(\lambda_H - \lambda_{\text{dV}}), \\ \partial_{v_Q} V_{\text{eff}} = 4C_v(\rho, \tau)v_Q(v_Q^2 - \mu_B(\rho, \tau)^2 J_v).$$

For $v_Q \neq 0$ and positive selector coefficients, stationarity gives (70). Equation (71) is the canonical radial relation from (68). $\square$

On the selector branch, the squared terms in V_{sel} vanish. Thus ρ and τ obey the bulk coefficient-balance equations

$$0 = 2B_R \tau^{-2} \rho^{-3} - 6B_H \tau^{-2} \rho^{-7} + 6B_0 \tau^{-4} \rho^5 + 2B_2 \tau^{-4} \rho \\ - 2B_4 \tau^{-4} \rho^{-3} - 6B_6 \tau^{-4} \rho^{-7}, \quad (72)$$

$$0 = 2B_R \tau^{-3} \rho^{-2} - 2B_H \tau^{-3} \rho^{-6} - 4B_0 \tau^{-5} \rho^6 - 4B_2 \tau^{-5} \rho^2 \\ - 4B_4 \tau^{-5} \rho^{-2} - 4B_6 \tau^{-5} \rho^{-6} - 3B_{\text{loc}} \tau^{-4}. \quad (73)$$

Example 2.2 (Selector branch as the direct scalar slice). On (70),

$$\frac{v_{\text{EW}}^2}{2\mu_B(\rho, \tau)^2} = 1 + \sqrt{3} = |X_-(2; 1)|.$$

Also,

$$\frac{M_H^2}{\mu_B(\rho, \tau)^2} = 4\lambda_{\text{dV}}(1 + \sqrt{3}) = \frac{3 + \sqrt{57}}{8} = \left| X_-\left(\frac{3}{4}; 1\right) \right|.$$

This is the direct boundary-scalar assignment from Part II written as a stationary branch of V_{eff} .

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